

Liam Campbell

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EDUCATION

Stanford University | Palo Alto CA

Masters of Science in Mechanical Engineering

Expected June 2026

- GPA: 4.00, Mechatronics and Automatic Controls Track
- Mechanical System Design Course Assistant; mentoring 90+ students over two quarters.
- Courses: Adv. Mechanical System Design, Adv. Dynamics & Simulation, Embedded Product Design, Control Design, Optimal Control, State Estimation, Convex Optimization, Design Optimization

Tufts University | Medford MA

Bachelor of Science in Mechanical Engineering

May 2024

- GPA: 3.96, Dean's List, Tau Beta Pi, Men's Ultimate Frisbee

PROJECTS

Mass Optimal Compliant Structure Design | *Stanford ME204*

Dec 2025

- Created a custom deformation model using first principles and finite beam analysis to within ~5% of FEA.
- Won lowest mass award using a local and global mass optimization study on a parameterized simple model.

Long Endurance Spring Cycling Mechanism | *Stanford ME204*

Dec 2025

- Eliminated 100% of torque fluctuations to maximize power with an optimized variable radius pulley profile.
- Boosted power output by 7.3% by implementing a no-load cooling phase through a mutilated gear / damper.
- Optimized mass, energy loss, and FOS over 2,932 high torque cycles to win highest cycle count award.

1st Place Mechatronics Robot | *Stanford ME218B*

Mar 2025

- Wrote low level embedded software in C to interface PIC32s, environmental sensors, motors & encoders.
- Designed low latency motor control at 250Hz using varying priority interrupt service routines (ISR).
- Architected a hierarchical state machine coordinating navigation and 2dof arm actuation across two PIC32s.

Space Arcade Game | *Stanford ME218A*

Dec 2024

- Wired 28 analog/digital I/O on a single constrained PIC32 using shift registers and an SPI daisy chain.
- Built a state machine to synchronously monitor inputs and control audible, haptic, and visual feedback.

WORK EXPERIENCE

Beta Technologies | *Manufacturing Engineering Intern*

Jun 2025 - Oct 2025

- Designed manufacturing tooling for 0.005" tolerance wing and fuselage subassembly from first principles.
- Produced 17 CNC / welded parts in 3DX with GD&T drawings and manufacturing instructions.
- Identified potential assembly line optimization of 89% using a custom MILP optimization program.

Center for Engineering Education & Outreach | *Undergraduate Researcher*

Jan 2024 - Aug 2024

- Created a human in the loop system for autonomous AI robotic programming with visual self-verification.
- Built a ROS-based Python package for 6DOF arm control, tailored for students and researchers.

Sol Clarity | *Greentown Labs* | *Embedded Engineering Intern*

Sep 2023 - Dec 2023

- Processed 10+ PCB with Python scripts to characterize and predict output of electrical systems in the field.
- Automated electrical tests via C++ firmware, speeding up data collection and increasing trial consistency.
- Used oscilloscope to troubleshoot and solder-repair custom high-voltage and logic level PCBs.

SKILLS

Software: Solidworks, Onshape, Catia, MatLab, LabView, KiCad, Excel, Adobe

Embedded / Code: Python, C, C++, Julia, Arduino, Raspberry Pi, Logic Analyzer, MPLAB, Git

Fabrication: Rapid Prototyping, 3D Printing, CNC, Mill, Lathe, PCB Design, Soldering, Oscilloscope, GD&T

Robotics / Controls: ROS2, PID (Bode, Root Locus, Nyquist), State Space Control, Kalman Filtering, PyBullet, PyTorch, CVXPY, Non-Linear Optimization